IEOR 4101: Probability, Statistics and Simulation

Lecture 8: Special Distributions and Random Variables

Discrete Uniform Random Variable

- A discrete uniform random variable X on {a, a+1, ..., b} is equally likely to be any x in {a, a+1, ..., b}

$\{1, 2, \dots, 6\}$

- E.g., fair die roll; fair coin toss

$\{0, 1\}$

- P(X = x) = 1/(b-a+1), x = a, a+1, ..., b

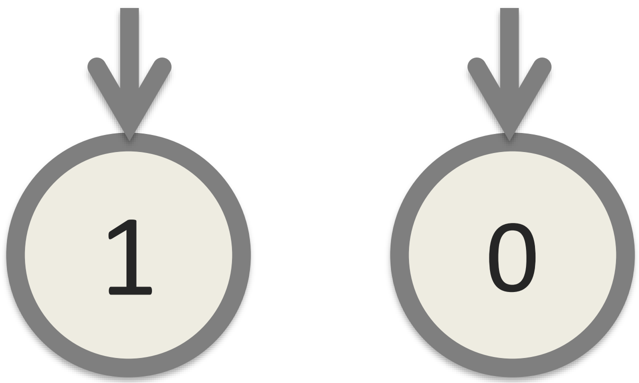
- E[X]; Var(X)

$$\mu = E[X] = \frac{1}{b-a+1}\left(a + (a+1) + \dots + b\right)$$

$$Var(X) = \frac{1}{b-a+1}\left((a-\mu)^2 + \dots + (b-\mu)^2\right)$$

Bernoulli Random Variables

- A Bernoulli random variable is used to indicate the success/failure of an event, with a success probability p; e.g., a coin toss
- Sometimes called Bernoulli trial
- $P(X = 1) = p$; $P(X = 0) = 1-p$



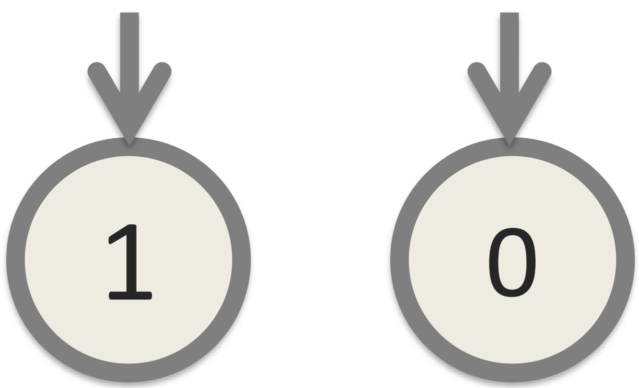
$$E[X] = 1 \times P(X=1) + 0 \times P(X=0) = P(X=1) = p$$

$$\mathrm{var}(X) = E[X^2] - (E[X])^2 = E[X] - (E[X])^2$$

$$= p - p^2 = p(1-p)$$

Bernoulli Random Variables

- A Bernoulli random variable is used to indicate the success/failure of an event, with a success probability p; e.g., a coin toss
- Sometimes called Bernoulli trial
- $P(X = 1) = p$; $P(X = 0) = 1\text{-}p$



- $E[X] = 1 \times P(X{=}1) + 0 \times P(X{=}0) = p$
- $Var(X) = E[X^2] – (E[X])^2 = p – p^2 = p(1 – p)$
- Notation: Ber(p)

Binomial Distribution

- Consider n independent repetitions of a Bernoulli trial with success probability p
- X denotes the number of successes in n trials
- Pmf: $P(X = k) = p(k) = \binom{n}{k} p^k(1-p)^{n-k}$, $k=0,\ldots,n$

Position of trials

① ② . . . ⓝ

pick k positions out of n positions.

This gives $\binom{n}{k}$ possibilities.

Express $X = X_1 + \cdots + X_n$ where

$X_i \sim Ber(p)$, independent.

$$E[X] = E[X_1] + \cdots + E[X_n] = np$$

$$var(X) = var(X_1 + \cdots + X_n)$$

$$= var(X_1) + \cdots + var(X_n)$$ because X_i's are independent

$$= np(1-p)$$

Binomial Distribution

- Consider n independent repetitions of a Bernoulli trial with success probability p
- X denotes the number of successes in n trials
- Pmf: $P(X = k) = p(k) = \binom{n}{k} p^k(1-p)^{n-k}$, $k=0,\ldots,n$
- $E[X]=np$
- $Var(X)=np(1-p)$
- Denoted by Bin(n, p)

Binomial Distribution

- **Example**: Disks produced by a certain company will be defective with probability 0.01 independently of each other. The company sells disks in packages of 10 and offers a money-back guarantee that at most 1 out of 10 disks is defective. What proportion of packages is returned? What is the proportion of packages having exactly two defective disks?

$$X = \# \text{ defective}$$

$$\sim Bin(10, 0.01)$$

$$P(\text{return}) = P(X > 1) = 1 - P(X=0) - P(X=1)$$

$$= 1 - (1-0.01)^{10} - 10 \times 0.01 \times (1-0.01)^{10-1}$$

$$P(X=2) = \binom{10}{2} 0.01^2 (1-0.01)^{10-2}$$

$$= \frac{10 \times 9}{2} 0.01^2 \ 0.99^8$$

Geometric Distribution

- X = # of Bernoulli trials with success probability p until we see a first success

Geometric Distribution

- X = # of Bernoulli trials with success probability p until we see a first success
- Pmf: $P(X = k) = (1-p)^{k-1}p$, $k = 1, 2, \ldots$
- $E[X] = 1/p$
- $Var(X) = (1-p)/p^2$
- Denoted by $Geo(p)$

Geometric Distribution

- **Example**: One die is rolled over and over. How many rolls are required so that the probability of rolling a six within this number is at least 50%?

Find k such that

$$P(X \leq k) \geq 50\%$$

X = # rolls needed to see "6"

$$X \sim Geo\left(\frac{1}{6}\right)$$

$$P(X \leq k) = 1 - P(X > k)$$

$$= 1 - (1-p)^k$$

because $X > k$ is
the same as
no success in the
first k trials

Find k such that

$$P(X \le k) \ge 50\%$$

or

$$1 - (1-p)^k \ge 50\%$$

$$50\% \ge (1-p)^k$$

$$\log \frac{1}{2} \ge k \log(1-p)$$

$$k \ge \frac{\log \frac{1}{2}}{\log(1-p)}$$

Geometric Distribution

- **Example**: One die is rolled over and over. How many rolls are required so that the probability of rolling a six within this number is at least 50%?

- **Solution**. Let this number be k. Let X be a Geo(1/6) random variable. We are interested in the value of k so that $P(X \leq k) \geq 0.5$

- Or $P(X > k) \leq 0.5$
- $P(X > k) = (1-p)^k = (5/6)^k \leq 0.5 \rightarrow k \geq 3.8$
- So smallest k is 4

Geometric Distribution

- **Example**. Tom and his mom lost each other while wandering through a crowded amusement park. Their iphones are out of battery. Tom's mom, being a wise woman, foresaw this and made a prior agreement with Tom. Tom will wait at the entrance of one of the 15 attractions and his mom searches the attractions in a random order (the "wait-for-mommy" strategy). What are the expected value and standard deviation of the number of searches before Tom's mom finds him?

X = # searches to find Tom

X take values $1, \ldots, 15$

$\Rightarrow$ X is not geometric

X is not binomial

because X cannot be 0.

X is uniform from 1 to 15.

Tom

1^{st} pick 2^{nd} pick

Think of mom picking an ordering of attractions. Then Tom is equally likely to be at any of the positions.

$$P(X = k) = \frac{1}{15} \quad \text{for } k = 1, \ldots, 15.$$

$E[X] = \ldots$

$\mathrm{var}(X) = \ldots$

standard deviation $= \sqrt{\mathrm{var}(X)} = \ldots$

Geometric Distribution

- **Example**. Tom and his mom lost each other while wandering through a crowded amusement park. Their iphones are out of battery. Tom's mom, being a wise woman, foresaw this and made a prior agreement with Tom. Tom will wait at the entrance of one of the 15 attractions and his mom searches the attractions in a random order (the "wait-for-mommy" strategy). What are the expected value and standard deviation of the number of searches before Tom's mom finds him?

- Suppose now that Tom and his mom agrees that, in intervals of 10 minutes, they will each pick a random entrance and go to their chosen entrances. What are the expected value and standard deviation of the number of searches in this case?

X = # searches needed.

$$X \sim Geo\left(\frac{1}{15}\right)$$

$$E[X] = 15$$

$$var(X) = \frac{1 - \frac{1}{15}}{\left(\frac{1}{15}\right)^2}$$

standard deviation $= \sqrt{var(X)} = \ldots$

Poisson Distribution

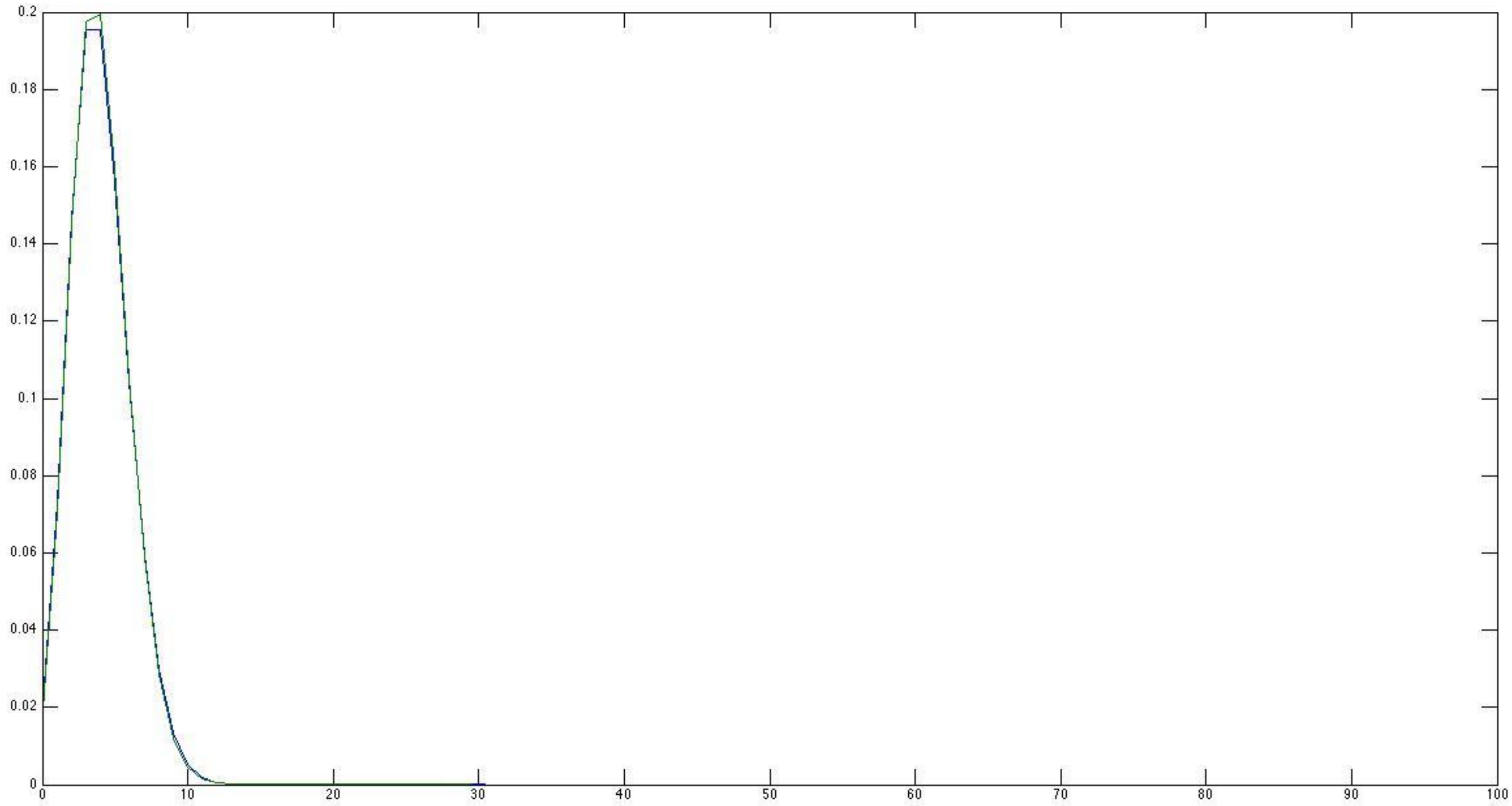
- Poisson r.v. counts the number of occurrences of interest in a segment of time or space; possible values are 0, 1, 2, …
- Poisson distribution with parameter $\lambda = np$ approximates Bin(n, p) when n is large and p is small; e.g., number of calls received by an emergency center
- X has a Poisson distribution with parameter $\lambda>0$ if its pmf satisfies

$$P(X = k) = e^{-\lambda}\lambda^k/k!, \ k = 0, 1, 2, \dots$$

- $E[X] = Var(X) = \lambda$
- Denoted by $Poi(\lambda)$

Poisson Distribution

- Plot shows Poi(λ) with λ = 4 and Bin(n, p) with n = 100, p = 0.04
- Blue is Poisson pmf and green is Binomial pmf



Poisson Distribution

- **Example**. At Whole Foods Grocery, customers are thought to arrive at the checkout section at a rate of **16 customers per hour**. The store manager believes that the service time for each customer is **constant at six minutes**. Suppose that during a **six minute interval, three checkers are available**, i.e., three customers can be served during a six-minute time segment. The manager is interested in the probability that one or more customers will have to wait for service during a six-minute window. Determine this probability, supposing that the number of arrivals in all time segment is a Poisson random variable.

X = # arrivals of customers during 6-min interval

P(at least one customer waits)

$= P(X \geq 4)$

$$X \sim Poi\left(16 \times \frac{6}{60}\right) = Poi(1.6)$$

$$P(X \geq 4)$$

$$= 1 - P(X=0) - P(X=1) - P(X=2) - P(X=3)$$

$$= 1 - \frac{e^{-\lambda}\lambda^0}{0!} - \frac{e^{-\lambda}\lambda^1}{1!} - \frac{e^{-\lambda}\lambda^2}{2!} - \frac{e^{-\lambda}\lambda^3}{3!}$$

$$= 1 - e^{-\lambda} - e^{-\lambda}\lambda - \frac{e^{-\lambda}\lambda^2}{2} - \frac{e^{-\lambda}\lambda^3}{6}$$

for $\lambda = 1.6$

Hypergeometric Distribution

- Hypergeometric distributions are often used in situations where we are sampling *without* replacement in a finite population

- Population has two mutually exclusive attributes
  - E.g., male vs female, positive vs negative to a blood test, etc

- Want to count the number in the sample with a certain attribute
  - E.g., number of male, number of positive reactions to a blood test, etc

Hypergeometric Distribution

- For a population of size N, suppose that there are K "successes"
  - N = 30 and K = 12 in previous example
- A random sample size of n
- X = number of successes in the sample
- P(X = k) = $\left[\binom{K}{k} \times \binom{N-K}{n-k}\right] / \binom{N}{n}$

# of ways to pick the k successes

# of ways to pick the n-k failures

# of ways to pick the sample

Parameters: N, K, n

Hypergeometric Distribution

- **Example (Gender equity)**. Suppose a company needs to downsize one department having 30 people – 12 women and 18 men. 10 people were laid off, and upper management said that the layoffs were done randomly. Of the 10 people, 8 were women. A labor attorney is interested in the probability of 8 or more women being laid off by chance

X = # women laid off

$P(X \geq 8)$

X is hypergeometric variable with

$N = 30$, $K = 12$, $n = 10$

$$P(X \geq 8)$$

$$= P(X = 8) + P(X = 9) + P(X = 10)$$

$$= \frac{\binom{12}{8}\binom{18}{2}}{\binom{30}{10}} + \frac{\binom{12}{9}\binom{18}{1}}{\binom{30}{10}} + \frac{\binom{12}{10}\binom{18}{0}}{\binom{30}{10}}$$

Hypergeometric Distribution

- **Example (Gender equity)**. Suppose a company needs to downsize one department having 30 people – 12 women and 18 men. 10 people were laid off, and upper management said that the layoffs were done randomly. Of the 10 people, 8 were women. A labor attorney is interested in the probability of 8 or more women being laid off by chance

- **Solution**. The probability of having more than 8 women by chance is

$$\frac{\binom{12}{8}\times\binom{18}{2}}{\binom{30}{10}} + \frac{\binom{12}{9}\times\binom{18}{1}}{\binom{30}{10}} + \frac{\binom{12}{10}}{\binom{30}{10}} = 0.0026$$

Hypergeometric Distribution

- When the population size N is large compared to the sample size n, we can approximate the hypergeometric distribution using a binomial distribution.

- In previous example, if the department has say 300 people, of which 120 are women and 180 are men. What is the probability of having 8 women in a random sample of size 10?

- Binomial approximation: probability of women = 120/300 = 0.4
- P(8 women) $\approx \binom{10}{8} \times 0.4^8 \times 0.6^2 = 0.01$
- Calculation using hypergeometric distribution gives 0.0097

hypergeometric:

- finite population
- draw n individuals without replacement
- # successes among the n individuals

binomial:

- infinite population
- run n trials
- # successes among the n trials

When the population size in a hypergeometric variable is very big, without replacement is almost the same as with replacement

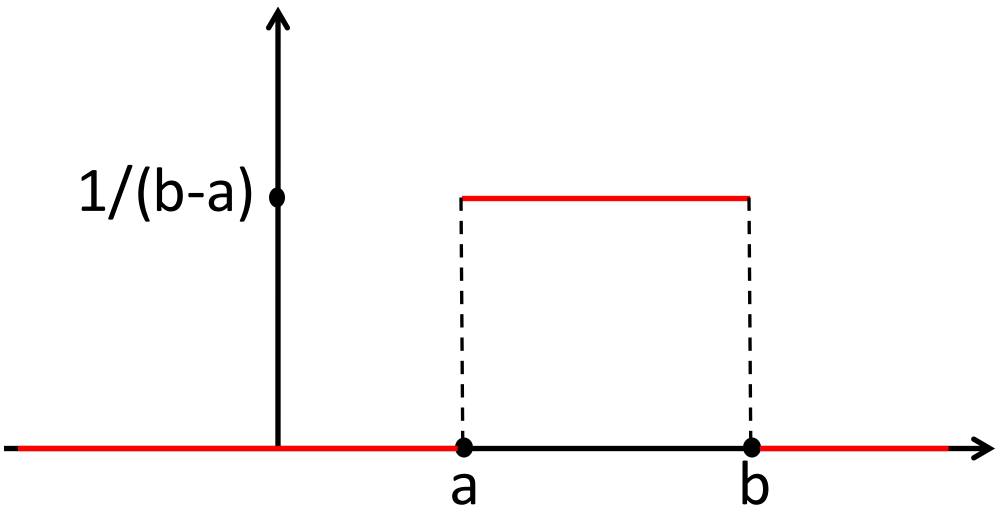
$\Rightarrow$ hypergeometric $\approx$ binomial with n &

with N, K, n $\qquad$ $p = \dfrac{K}{N}$

Uniform Random Variables

- A random variable X is uniformly distributed over [a, b] if its probability density function (pdf) f(x) is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$



- Denote it by Unif(a, b)
- Used to model a "completely random" selection over the interval

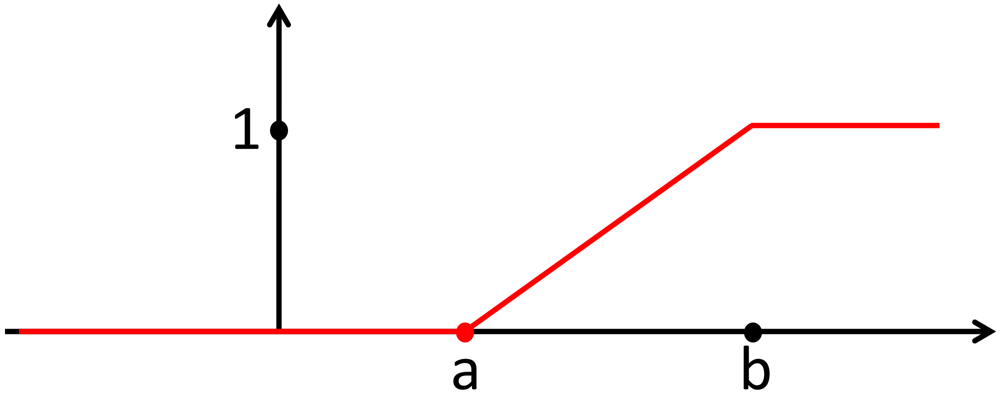
Uniform Random Variables

- Note that this is a valid pdf because

$$\int_a^b \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b dx = \frac{1}{b-a} \cdot (b-a) = 1$$

- The (cumulative) distribution function of Unif(a, b) is

$$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \le x \le b \\ 1 & \text{if } x > b. \end{cases}$$



Uniform Random Variables

- Expectation of X ~ Unif(a, b) is

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b-a} dx = \left.\frac{x^2}{2(b-a)}\right|_a^b = \frac{a+b}{2}$$

- Variance of X can be computed as follows

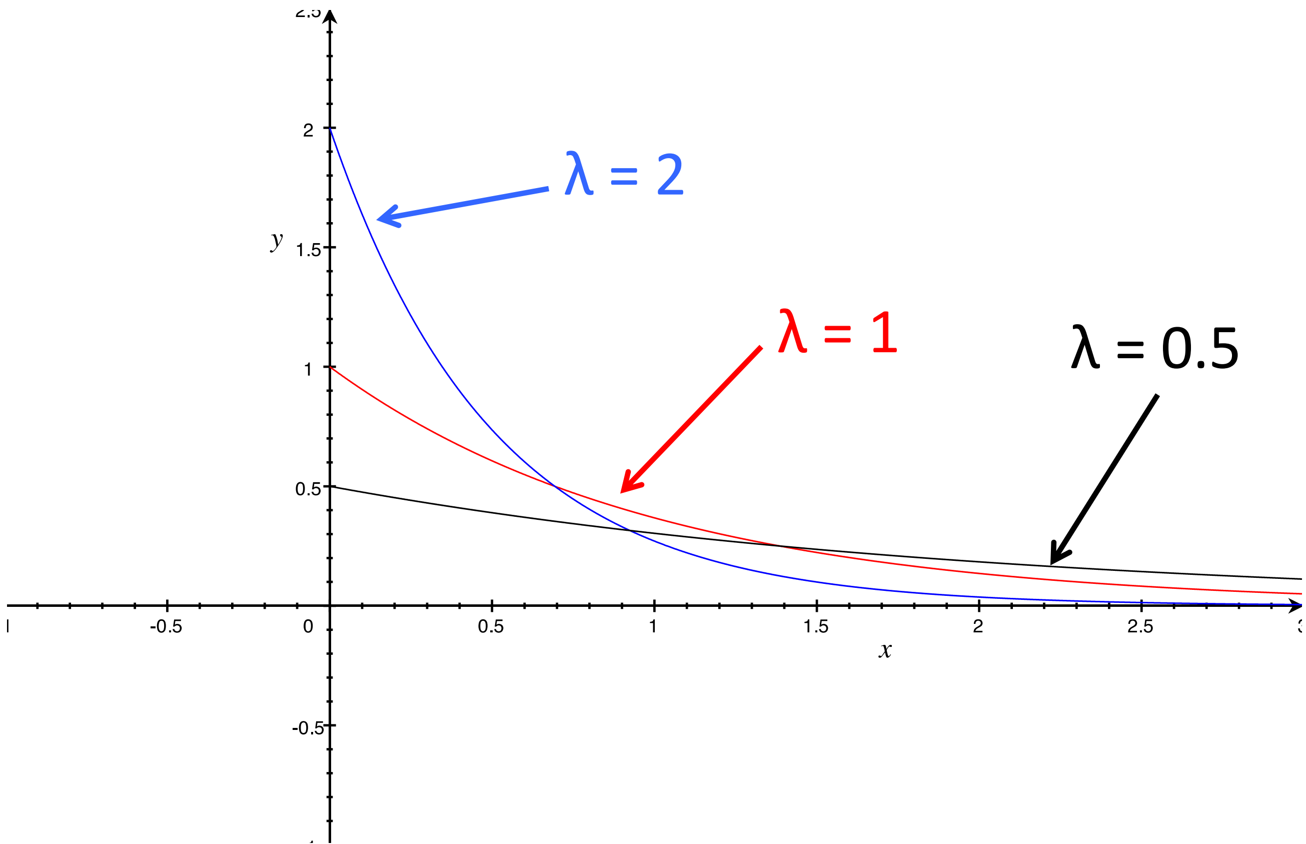
$$\mathbb{E}[X^2] = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \left.\frac{x^3}{3(b-a)}\right|_a^b = \frac{b^3-a^3}{3(b-a)}$$

$$\begin{aligned} Var(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{b^3-a^3}{3(b-a)} - \left(\frac{a+b}{2}\right)^2 \\ &= \frac{b^2+ab+a^2}{3} - \frac{a^2+2ab+b^2}{4} = \frac{(b-a)^2}{12} \end{aligned}$$

Exponential Random Variables

- A random variable X has an *exponential* pdf with parameter λ > 0 if the pdf f(x) is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$



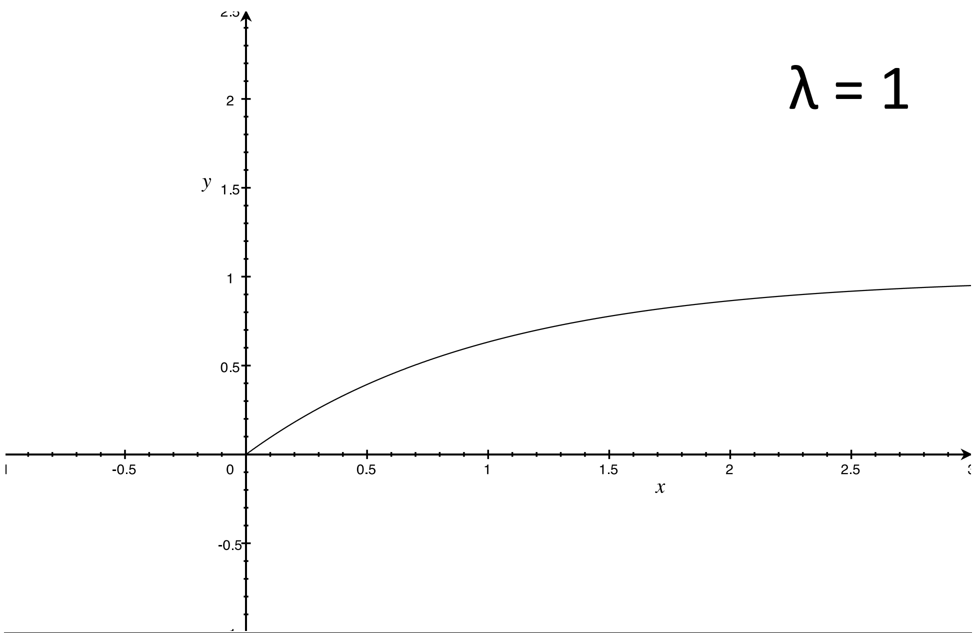
Exponential Random Variables

- Denoted by Exp(λ)
- Often used to model the time till some event (e.g., earthquake, phone call, radioactive decay, etc) occurs
- cdf F(x) is given by

$$F(x) = \int_0^x \lambda e^{-\lambda y} dy = -e^{-\lambda y}\Big|_0^x = 1 - e^{-\lambda x}, x \geq 0$$

- E[X] = 1/λ
- Var(X) = $1/\lambda^2$

λ = 1



Exponential Random Variables

- **Memoryless property**. Let X be an exponential rv. Then for any positive s and t, $P(X - s > t \mid X > s) = P(X > t)$
- In words: think of X as the time till some event occurs. I have observed for s amount of time and don't see the occurrence of the event. Conditioned on this, the random amount of time that I have to wait *further* (i.e., X-s) till the occurrence has the same distribution as X

Exponential Random Variables

- Proof of memoryless property:

$$\begin{aligned}
\mathbb{P}(X - s > t \mid X > s) &= \frac{\mathbb{P}(X > s + t, X > s)}{\mathbb{P}(X > s)} \\
&= \frac{\mathbb{P}(X > s + t)}{\mathbb{P}(X > s)} \\
&= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}
\end{aligned}$$

Exponential Random Variables

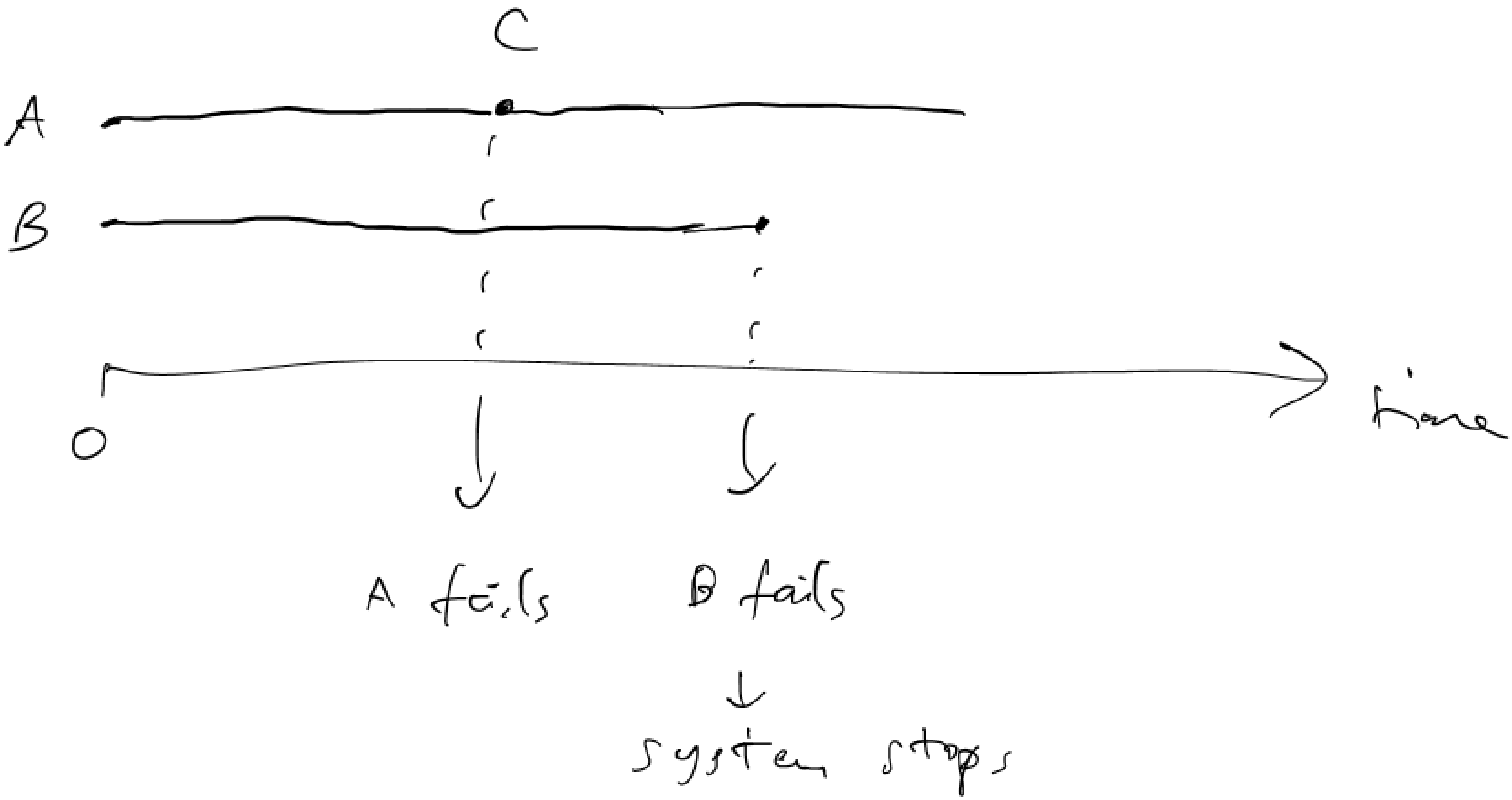
- **Property 2**: Suppose $X_1 \sim Exp(\lambda_1)$, $X_2 \sim Exp(\lambda_2)$, …, $X_n \sim Exp(\lambda_n)$ are independent. Let $Y = \min\{X_1, X_2, \ldots, X_n\}$. Then $Y \sim Exp(\lambda_1 + \lambda_2 + \ldots + \lambda_n)$.

- Proof. For any s > 0,

$$\begin{aligned}
\mathbb{P}(Y > s) &= \mathbb{P}(\min\{X_1, X_2, \cdots, X_n\} > s) \\
&= \mathbb{P}(X_1 > s, X_2 > s, \cdots, X_n > s) \\
&= \mathbb{P}(X_1 > s)\mathbb{P}(X_2 > s) \cdots \mathbb{P}(X_n > s) \\
&= e^{-\lambda_1 s} \cdots e^{-\lambda_n s} \\
&= e^{-(\lambda_1 + \cdots + \lambda_n)s}
\end{aligned}$$

Exponential Random Variables

- **Example**. A crew of workers have 3 interchangeable machines A, B and C. They need 2 of them to be functional simultaneously to do their job. The lifetime of the machines are independently distributed as Exp(λ). They start A and B first and keep C as reserve to replace whichever of A or B that breaks down first. They then continue until only one of the machines remains functional, at which time they are forced to stop. What is the expected time to stop? What is the probability that at this time, the only functional machine is C?


C
A
B
0
time
A fails
B fails
system stops

$$E[\underbrace{\text{time for the system to stop}}]$$

$$= \overbrace{\text{time for either A or B to fail}}^{= T_1}$$

$$+ \underbrace{\text{time for either C or the remaining machine among A or B to fail}}_{= T_2}$$

$$T_1 = \min\{T_A, T_B\}$$

where

T_A = failure time of A

T_B = failure time of B

$$T_A, T_B \sim Exp(\lambda)$$

$T_1 \sim Exp(\lambda + \lambda) = Exp(2\lambda)$ by 2^{nd} property

By memoryless property, the survivor of A or B at time T_1 is still $Exp(\lambda)$.

Call this remaining time to failure for this survivor $\tilde{T} \sim Exp(\lambda)$

$T_2 = \min\{\tilde{T}, T_C\}$ where T_C = failure time of C $\sim Exp(\lambda)$

$$T_2 \sim \mathrm{Exp}(\lambda + \lambda) = \mathrm{Exp}(2\lambda)$$

by 2^{nd} property

$$E[T_1 + T_2] = E[T_1] + E[T_2]$$

$$= \frac{1}{2\lambda} + \frac{1}{2\lambda}$$

$$= \frac{1}{\lambda}$$

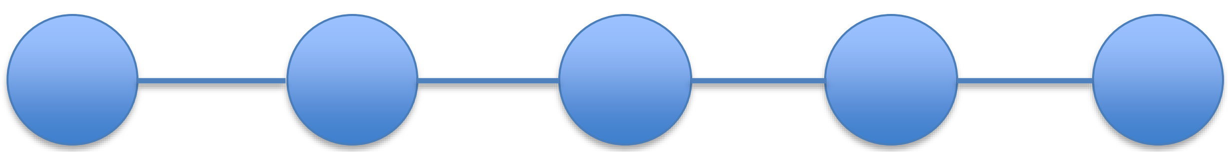
When A or B fails, the remaining time to fail for the survivor $\sim \mathrm{Exp}(\lambda)$ by memoryless property. This survivor "$\tilde{T}$"

competes with C whose failure time $\sim \mathrm{Exp}(\lambda)$. "$T_C$"

$P(\tilde{T} < T_C)$ must be $\frac{1}{2}$ by symmetry.

Exponential Random Variables

- **Example**. A series system has 5 components. The system is functional if and only if *all* of its components are functional. Suppose that the lifetimes of the components are independent exponential random variables with mean 10 minutes each. What is the probability that the system survives for 2 minutes?



$X_1, \ldots, X_5 \sim Exp(\frac{1}{10})$

$\min\{X_1, \ldots, X_5\} \sim Exp(\frac{5}{10}) = Exp(\frac{1}{2})$

$P(\min\{X_1, \ldots, X_5\} > 2)$

$= e^{-2 \cdot \frac{1}{2}} = e^{-1}$

Exponential Random Variables

- **Example**. A series system has 5 components. The system is functional if and only if *all* of its components are functional. Suppose that the lifetimes of the components are independent exponential random variables with mean 10 minutes each. What is the probability that the system survives for 2 minutes?

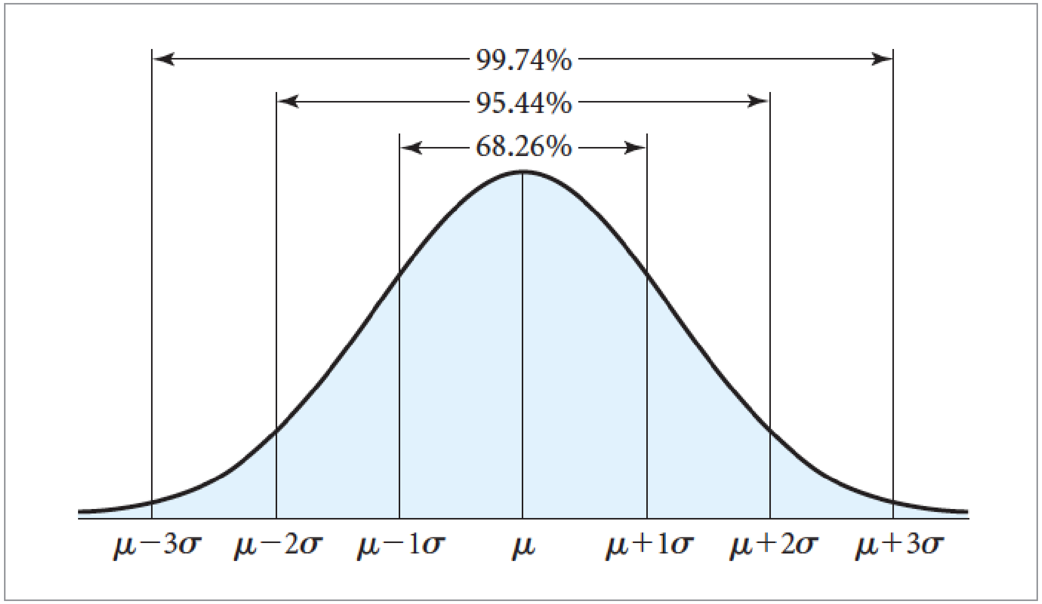
- **Solution**. Let X_i be the lifetime of the i^{th} component, i=1, 2, …, 5. Then, X_1, X_2, …, X_5 are independent exponential random variables with parameter 1/10 each. The failure time of the system is $Y = \min\{X_1, X_2, \ldots, X_5\}$, an exponential random variable with parameter $5 \times 1/10 = 1/2$ by **Property 2**. Thus, the desired probability is

$$\mathbb{P}(Y \geq 2) = \int_2^\infty \frac{1}{2} e^{-\frac{x}{2}} dx = -e^{-\frac{x}{2}} \;|_{x=2}^{\infty} = e^{-1}$$

Normal Random Variables

- A bell-shaped pdf curve
- Random variable X is normally distributed with mean μ and variance σ^2 if it has pdf

$$f(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, x \in \mathbb{R}$$



Normal Random Variables

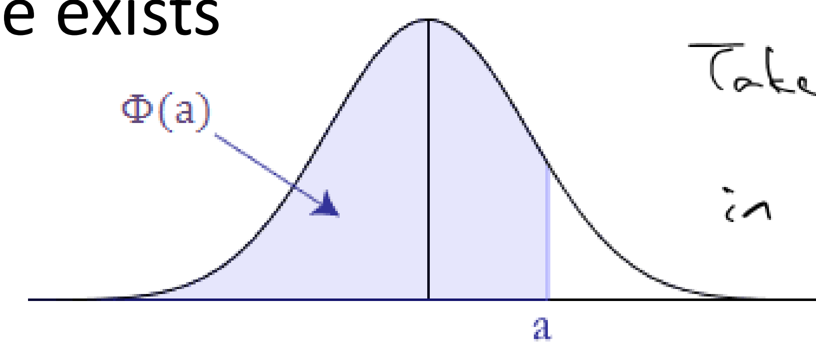
- Normal distribution and r.v. show up in many natural phenomena and are important in statistics (later)
- $E[X] = \mu$, $Var(X) = \sigma^2$
- Denote it by $N(\mu, \sigma^2)$
- $N(0, 1)$ is called the standard normal distribution

Normal Random Variables

- **Property 1**: If X~N(μ, σ^2), then aX+b~N(aμ+b, $a^2\sigma^2$)
- **Property 1'**: If X~N(μ, σ^2), then Z = (X-μ)/σ ~ N(0, 1)
- Moral: all normal rv can be "reduced" to the standard normal
- Often use Φ(x) to denote the cdf of standard normal

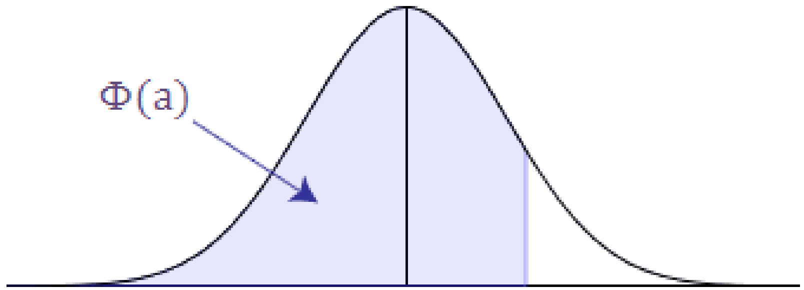
$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy$$

- No closed-form formula for Φ(x) but the standard normal table exists



Take $a = \frac{1}{\sigma}$, $b = -\frac{\mu}{\sigma}$, in Property 1 $\Rightarrow$ get Property 1'.

Normal Random Variables



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9031	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

Normal Random Variables

- **Example**. After completing a study, the Chicago O’Hare Airport managers have concluded that the time needed to get passengers loaded onto an airplane is (approximately) normally distributed with mean 15 minutes and standard deviation 3.5 minutes. What is the probability that a flight will take at least 22 minutes to get a passenger loaded?

X = time to load passenger

$\sim N(15, 3.5^2)$

$Z \sim N(0,1)$

$$P(X \geq 22) = P\left(\frac{X-15}{3.5} \geq \frac{22-15}{3.5}\right) = P(Z \geq 2)$$

$$= 1 - \Phi(2) = 0.9772$$

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- **Solution**. Let X be the random time. Then $X \sim N(15, 3.5^2)$. So, $(X-15)/3.5 \sim N(0, 1)$.
- $P(X \geq 22) = P\left(\frac{X-15}{3.5} \geq \frac{22-15}{3.5}\right) = P(Z \geq 2) = 1 - \Phi(2)$
- $\Phi(2) = 0.9772$; so $P(X \geq 22) = 1 - 0.9772 = 0.0228$

Normal Random Variables

- **Property 2**: if $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent normal rv, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

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- **Example**. Data from the National Oceanic and Atmospheric Administration indicate that the yearly precipitation in Los Angeles is a normal rv with mean 12.08 inches and standard deviation 3.1 inches. What is the probability that the total precipitation during the next 2 years will exceed 30 inches? Assume precipitations are independent across years.

X_1 = yearly precipitation for 1^{st} yr

$\sim N(12.08, 3.1^2)$

X_2 = . . . 2^{nd} . .

$\sim N(12.08, 3.1^2)$

X_1, X_2 are independent.

$$P(X_1 + X_2 > 30)$$

$$X_1 + X_2 \sim N(12.08 \times 2,\ 3.1^2 \times 2)$$

$$= P\left(\frac{X_1 + X_2 - 12.08 \times 2}{\sqrt{3.1^2 \times 2}} > \frac{30 - 12.08 \times 2}{\sqrt{3.1^2 \times 2}} \right)$$

$$= P(Z > 1.33) \qquad Z \sim N(0, 1)$$

$$= 1 - \Phi(1.33) = 1 - 0.9082$$

Normal Random Variables

- **Property 2**: if $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent normal rv, then $X_1+X_2 \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$

- **Example**. Data from the National Oceanic and Atmospheric Administration indicate that the yearly precipitation in Los Angeles is a normal rv with mean 12.08 inches and standard deviation 3.1 inches. What is the probability that the total precipitation during the next 2 years will exceed 30 inches? Assume precipitations are independent across years.
- **Solution**. Let X_1 and X_2 be the precipitations for the next 2 years. $X_1, X_2 \sim N(12.08, 3.1^2)$ are independent. Then $X_1+X_2 \sim N(2\times 12.08, 2\times 3.1^2)$.
- $P(X_1+X_2 > 30) = P\left(\frac{X_1+X_2-2\times 12.08}{\sqrt{2\times 3.1^2}} > \frac{30-2\times 12.08}{\sqrt{2\times 3.1^2}}\right) = P(Z > 1.33) = 1 - \Phi(1.33) = 1 - 0.9082 = 0.0918$